

PY1010 — Physics for Biological and Chemical Sciences

Summer 2024 Examination · Paper 1 · Full Marking Scheme

Module	PY1010 — Physics for the Biological and Chemical Sciences
Internal examiners	Prof. P. Callanan, Prof. A. Ruth
Institution	University College Cork
Session	Summer 2024, Paper 1
Duration	180 minutes (+ 15 minutes reading time)
Structure	Six questions, each given as an either/or choice (A or B). Answer one option of each of the six questions and all parts of that option. Marks for each part are shown in square brackets.
Materials	Non-programmable calculator permitted; constants on the paper's "useful data" page

How to use this scheme. Every answer is worked out in small numbered steps, with a "Rules you need" box listing the formulas used. Nothing is skipped, so you can see where every number comes from: every unit conversion, every substitution, every check. Read the answer first, then follow the steps.

Coverage note. This scheme works through **both options (A and B) of all six questions** in full. On the day you only choose one option per question, but working both lets you pick the one you're strongest on.

Source basis. Every answer is built **only** from the PY1010 lecture material given to students: the lecture slides (mechanics, fluids, thermodynamics) and the handwritten lecture notes (gas laws, electricity, magnetism, induction, waves and optics). No textbook or outside material is used, except where a highlighted note says the lecture notes don't cover something (Q3(A)(b), (d) and (e)).

Own-figure rule. Where a part carries forward a number from an earlier part, an answer that uses the correct method on an earlier wrong value still earns the method marks for the later part.

Numbers. $g = 9.81 \text{ m s}^{-2}$ and other constants from the paper's useful-data page. Intermediate values are kept unrounded; final answers are given to 2–3 significant figures. Answers differing only through rounding get full marks.

Study-Unit

Question 1 · Option A

Unit conversion, density, estimating numbers and kinetic energy of red blood cells — 10 marks

Given: Mean RBC volume $95 \mu\text{m}^3$; RBC density 1125 kg m^{-3} ; blood volume 4.7 litres; 4.8×10^6 RBCs per mm^3 of blood; 1 litre = 10^{-3} m^3 ; $1 \mu\text{m} = 10^{-6} \text{ m}$.

1(A)(a)

[2 marks]

How many μm^3 in 1 m^3 ?

ANSWER

$$1 \text{ m}^3 = 10^{18} \mu\text{m}^3$$

RULES YOU NEED

- $1 \text{ m} = 10^6 \mu\text{m}$ (micro = 10^{-6})
- $(a^m)^n = a^{m \times n}$

WORKING — STEP BY STEP

1 Start from the length conversion

$$1 \mu\text{m} = 10^{-6} \text{ m}, \text{ so } 1 \text{ m} = 10^6 \mu\text{m}$$

2 A cubic metre is a cube 1 m on each side

Each side is $10^6 \mu\text{m}$ long, so the volume is $(10^6 \mu\text{m}) \times (10^6 \mu\text{m}) \times (10^6 \mu\text{m})$:

$$1 \text{ m}^3 = (10^6 \mu\text{m})^3$$

3 Cube the power of ten

Multiply the powers: $6 \times 3 = 18$:

$$1 \text{ m}^3 = 10^{18} \mu\text{m}^3$$

EXAM TECHNIQUE

A very common slip is writing $10^6 \mu\text{m}^3$. The conversion factor must be cubed along with the unit: $(10^6)^3 = 10^{18}$.

Marks

Awarded for

1 Uses $1 \text{ m} = 10^6 \mu\text{m}$ and cubes it.

1 10^{18} .

1(A)(b)**[2 marks]**

Estimate the mass of a red blood cell.

ANSWER

$$m \approx 1.1 \times 10^{-13} \text{ kg}$$

RULES YOU NEED

- Density $\rho = m / V$, so $m = \rho V$
- $1 \mu\text{m}^3 = 10^{-18} \text{ m}^3$ (from part (a))

WORKING — STEP BY STEP**1 Convert the volume to m^3** Use part (a): each μm^3 is 10^{-18} m^3 :

$$V = 95 \mu\text{m}^3 = 95 \times 10^{-18} \text{ m}^3 = 9.5 \times 10^{-17} \text{ m}^3$$

2 Use $m = \rho V$

$$m = 1125 \text{ kg m}^{-3} \times 9.5 \times 10^{-17} \text{ m}^3$$

3 Multiply $1125 \times 9.5 = 10\,687.5$, so:

$$m = 10\,687.5 \times 10^{-17} \text{ kg} = 1.07 \times 10^{-13} \text{ kg}$$

Marks	Awarded for
1	Volume converted to m^3 and $m = \rho V$ used.
1	Correct mass with unit.

1(A)(c)**[3 marks]**

Estimate the total number of red blood cells in an average woman's body.

ANSWER

$$N \approx 2.3 \times 10^{13} \text{ red blood cells}$$

RULES YOU NEED

- 1 litre = 10^{-3} m^3
- 1 m = 10^3 mm , so $1 \text{ m}^3 = 10^9 \text{ mm}^3$
- Total = (cells per mm^3) $\times$ (number of mm^3)

WORKING — STEP BY STEP**1 Convert the blood volume to m^3**

$$4.7 \text{ L} = 4.7 \times 10^{-3} \text{ m}^3$$

2 Convert m^3 to mm^3

$$1 \text{ m} = 1000 \text{ mm}, \text{ so } 1 \text{ m}^3 = (10^3)^3 = 10^9 \text{ mm}^3:$$

$$4.7 \times 10^{-3} \text{ m}^3 \times 10^9 \text{ mm}^3/\text{m}^3 = 4.7 \times 10^6 \text{ mm}^3$$

3 Multiply by the cells in each mm^3

$$N = 4.8 \times 10^6 \text{ cells/mm}^3 \times 4.7 \times 10^6 \text{ mm}^3$$

4 Calculate

$$4.8 \times 4.7 = 22.56 \text{ and } 10^6 \times 10^6 = 10^{12}:$$

$$N = 22.56 \times 10^{12} = 2.3 \times 10^{13}$$

Marks	Awarded for
1	Litres converted to m^3 .
1	m^3 converted to mm^3 ($\times 10^9$).
1	$N \approx 2.3 \times 10^{13}$.

1(A)(d)**[3 marks]**

Estimate the total kinetic energy of all red blood cells when the average velocity of the blood flow is 8 mm/s.

ANSWER

$$E_{\text{kin}} \approx 7.7 \times 10^{-5} \text{ J}$$

RULES YOU NEED

- $E_{\text{kin}} = \frac{1}{2} m v^2$
- Total mass = number $\times$ mass of one
- 1 mm = 10^{-3} m

WORKING — STEP BY STEP**1 Total mass of all the red blood cells**

Multiply the number from (c) by the mass of one from (b):

$$M = 2.256 \times 10^{13} \times 1.069 \times 10^{-13} \text{ kg} = 2.41 \text{ kg}$$

2 Convert the speed to m/s

$$v = 8 \text{ mm/s} = 8 \times 10^{-3} \text{ m/s} = 0.008 \text{ m/s}$$

3 Square the speed

$$v^2 = (8 \times 10^{-3})^2 = 64 \times 10^{-6} = 6.4 \times 10^{-5} \text{ m}^2/\text{s}^2$$

4 Apply the kinetic energy formula

$$E_{\text{kin}} = \frac{1}{2} \times 2.41 \times 6.4 \times 10^{-5} = 7.7 \times 10^{-5} \text{ J}$$

Marks	Awarded for
1	Total RBC mass ≈ 2.4 kg (own-figure from (b), (c) accepted).
1	Speed in m s^{-1} and $E_{\text{kin}} = \frac{1}{2}mv^2$ used.
1	Correct value $\approx 7.7 \times 10^{-5}$ J.

Question 1(A) Total: 10 Marks

Question 1 · Option B

Friction rules, deceleration of a sliding penguin, coefficient of kinetic friction — 10 marks

Given: 12 kg penguin sliding on horizontal ice, initial speed 5.5 m s^{-1} , comes to rest after 15 m.

1(B)(a)

[2 marks]

An object rests on a surface. Which of statements (A)–(E) are false? Explain.

(A) Friction always opposes the motion. (B) Friction is independent of the normal force. (C) Friction is material dependent. (D) μ_s is always larger than μ_k . (E) The unit of μ_s and μ_k is $[\text{N m}^{-1}]$.

ANSWER

(B) and (E) are false. Friction is proportional to the normal force, and μ has no unit.

RULES YOU NEED

- Static: $F_{\text{friction}} \leq \mu_s N$. Kinetic: $F_{\text{friction}} = \mu_k N$
- Friction points opposite to the motion
- $\mu_k < \mu_s$; μ has no unit

WORKING — STEP BY STEP

1 Check (A)

The lecture rule: frictional forces always point opposite to the direction of motion.
True.

2 Check (B)

$F_{\text{friction}} = \mu N$: friction is **proportional** to the normal force, not independent of it.
False.

3 Check (C)

Different materials have different μ values (e.g. glass on glass vs Teflon on steel).
True.

4 Check (D)

The lecture table states $\mu_k < \mu_s$. **True.**

5 Check (E)

$\mu = F_{\text{friction}}/N$ is a force divided by a force: $N \div N$, so it has **no unit**. **False.**

Marks	Awarded for
1	(B) identified as false: friction $\propto$ normal force.
1	(E) identified as false: μ is unitless.

1(B)(b)(i)**[3 marks]**

Calculate the magnitude of the acceleration experienced by the penguin and show that it is $\approx 1 \text{ m s}^{-2}$.

ANSWER

$$|a| = 1.01 \text{ m s}^{-2} \approx 1 \text{ m s}^{-2}$$

RULES YOU NEED

- Constant acceleration: $v^2 = v_0^2 + 2a(x - x_0)$

WORKING — STEP BY STEP**1 List what you know**

Start speed $v_0 = 5.5 \text{ m/s}$, final speed $v = 0$ (it stops), distance $x - x_0 = 15 \text{ m}$. Time is not given.

2 Pick the equation without time

$$v^2 = v_0^2 + 2a(x - x_0)$$

3 Substitute

$$0 = 5.5^2 + 2 \times a \times 15 \Rightarrow 0 = 30.25 + 30a$$

4 Solve for a

$$a = -30.25 \div 30 = -1.008 \text{ m s}^{-2}$$

The minus sign means it is slowing down. The size is $1.008 \approx 1 \text{ m s}^{-2}$, as required.

Marks	Awarded for
1	Correct choice of equation, $v^2 = u^2 + 2as$.
1	Correct substitution with $v = 0$.
1	$ a \approx 1 \text{ m s}^{-2}$ shown.

1(B)(b)(ii)**[2 marks]**

What is the magnitude of the frictional force bringing the penguin to a halt?

ANSWER

$F \approx 12 \text{ N (12.1 N)}$

RULES YOU NEED

- Newton's second law: $F = m a$

WORKING — STEP BY STEP**1 Friction is the net force**

On flat ice, friction is the only horizontal force, so it causes the whole deceleration.

2 Apply $F = ma$

$$F = 12 \text{ kg} \times 1.008 \text{ m s}^{-2} = 12.1 \text{ N}$$

(Using $a = 1 \text{ m s}^{-2}$ exactly gives 12 N, also fine.)

Marks	Awarded for
1	$F = ma$ used.
1	$F \approx 12 \text{ N}$ (using $a = 1 \text{ m s}^{-2}$ gives exactly 12 N, also full marks).

1(B)(b)(iii)**[3 marks]**

What is the coefficient of kinetic friction μ_k between the penguin and the ice?

ANSWER

$$\mu_k \approx 0.10 \text{ (no unit)}$$

RULES YOU NEED

- Kinetic friction $F = \mu_k N$
- Flat surface: $N = mg$

WORKING — STEP BY STEP**1 Normal force**

$$N = m g = 12 \times 9.81 = 117.7 \text{ N}$$

2 Rearrange $F = \mu_k N$

$$\mu_k = F / N$$

3 Substitute

$$\mu_k = 12.1 \div 117.7 = 0.103$$

Force $\div$ force, so μ_k has no unit.

4 Shortcut check

$$\mu_k = ma/(mg) = a/g = 1.008 \div 9.81 = 0.103 \checkmark$$

Marks	Awarded for
1	Normal force $N = mg$.
1	$\mu_k = F/N$.
1	$\mu_k \approx 0.10$, unitless (own-figure from (ii) accepted).

Question 1(B) Total: 10 Marks

Question 2 · Option A

Mass vs weight, kinematics of Jules Verne's cannon — 10 marks

Given: Escape velocity 11 km s^{-1} ; cannon length 270 m ; maximum tolerable acceleration 50 g .

2(A)(a)

[3 marks]

Describe the difference between the mass and weight of an object, giving the SI unit of each.

ANSWER

Mass = amount of matter/inertia, fixed, in kg. Weight = gravitational force $w = mg$, depends on g , in N.

RULES YOU NEED

- Weight $w = mg$
- $g = G M_E / r^2$ depends on where you are

WORKING — STEP BY STEP

1 Mass

Mass is the amount of matter in an object, and measures its inertia (how hard it is to speed up or slow down). It is the same everywhere. SI unit: **kilogram (kg)**.

2 Weight

Weight is the force of gravity on the object: $w = mg$. It is a force, so its SI unit is the **newton (N)**.

3 The key difference

Weight depends on g , which changes with location. On the Moon your weight is about one sixth of that on Earth, but your mass is unchanged.

Marks	Awarded for
1	Mass correctly described (amount of matter / inertia), unit kg.
1	Weight as a force, $w = mg$, unit N.
1	Weight depends on g /location, mass does not.

2(A)(b)(i)**[1 mark]**Compute the escape velocity in units of m s^{-1} .**ANSWER**

$$v = 1.1 \times 10^4 \text{ m s}^{-1} \text{ (11 000 m s}^{-1}\text{)}$$

RULES YOU NEED

- 1 km = 1000 m

WORKING — STEP BY STEP**1 Convert kilometres to metres**

$$11 \text{ km/s} = 11 \times 1000 \text{ m/s} = 11\,000 \text{ m/s} = 1.1 \times 10^4 \text{ m s}^{-1}$$

Marks**Awarded for****1**11 000 m s^{-1} .

2(A)(b)(ii)**[5 marks]**

If the cannon had a length of 270 m, calculate the acceleration of the passengers whilst in the cannon (constant acceleration, reaching escape velocity as they leave).

ANSWER

$$a \approx 2.2 \times 10^5 \text{ m s}^{-2}$$

RULES YOU NEED

- Constant acceleration: $v^2 = u^2 + 2as$
- Use this form when time is not given

WORKING — STEP BY STEP**1 List the knowns**

Start from rest: $u = 0$. Final speed $v = 1.1 \times 10^4 \text{ m/s}$. Distance $s = 270 \text{ m}$. We want a . Time is not given.

2 Pick the right equation

The only kinematic equation without t is:

$$v^2 = u^2 + 2 a s$$

3 Rearrange for a

With $u = 0$: $v^2 = 2as$, so:

$$a = v^2 / (2 s)$$

4 Square the speed

$$v^2 = (1.1 \times 10^4)^2 = 1.21 \times 10^8 \text{ m}^2/\text{s}^2$$

5 Divide

$$a = 1.21 \times 10^8 \div (2 \times 270) = 1.21 \times 10^8 \div 540 = 2.24 \times 10^5 \text{ m s}^{-2}$$

Marks	Awarded for
1	Knowns identified: $u = 0$, v , s .
1	Correct equation chosen: $v^2 = u^2 + 2as$.
1	Rearranged for a .
1	Correct substitution in SI units.
1	$a \approx 2.2 \times 10^5 \text{ m s}^{-2}$.

2(A)(b)(iii)**[1 mark]**

Briefly discuss whether your answer to (ii) is feasible, if the maximum tolerable acceleration is $50 \times g$.

ANSWER

Not feasible: $\approx 23\,000\text{ g}$, far above the 50 g (490 m s^{-2}) the body can tolerate

RULES YOU NEED

- Express an acceleration in “g”s by dividing by 9.8 m s^{-2}

WORKING — STEP BY STEP**1 Work out the limit**

$$50\text{ g} = 50 \times 9.8 = 490\text{ m s}^{-2}$$

2 How many g is the cannon?

$$2.24 \times 10^5 \div 9.8 \approx 22\,900\text{ g}$$

3 Compare

$22\,900\text{ g}$ is about 460 times the 50 g limit. The passengers would not survive: **not feasible**.

Marks	Awarded for
1	Compares with 490 m s^{-2} and concludes not feasible.

Question 2(A) Total: 10 Marks

Question 2 · Option B

Centripetal acceleration and force in the Bohr model of hydrogen — 10 marks

Given: Electron mass 9.1×10^{-31} kg; orbit radius 5.3×10^{-11} m; 6.6×10^{15} orbits per second.

2(B)(a)

[2 marks]

For uniform circular motion write down the expressions for centripetal acceleration and centripetal force.

ANSWER

$$a_c = v^2/r; F_c = mv^2/r$$

RULES YOU NEED

- Centripetal force (from the lectures): $F = mv^2/r$
- Newton's second law: $a = F/m$
- $v = \omega r$

WORKING — STEP BY STEP

1 Centripetal force

The force needed to keep a mass m moving in a circle of radius r at speed v :

$$F_c = m v^2 / r (= m \omega^2 r)$$

2 Centripetal acceleration

Divide by m ($a = F/m$):

$$a_c = v^2 / r (= \omega^2 r)$$

Both point towards the centre of the circle.

Marks	Awarded for
1	$a_c = v^2/r$ (or $\omega^2 r$).
1	$F_c = mv^2/r$ (or $m\omega^2 r$).

2(B)(b)(i)**[2 marks]**

Calculate the total distance travelled by the electron per orbit.

ANSWER **3.3×10^{-10} m per orbit****RULES YOU NEED**

- Circumference of a circle = $2\pi r$

WORKING — STEP BY STEP**1 One orbit = one circumference**

$$C = 2 \pi r$$

2 Substitute

$$C = 2 \times \pi \times 5.3 \times 10^{-11} = 3.33 \times 10^{-10} \text{ m}$$

Marks	Awarded for
1	Circumference $2\pi r$.
1	3.3×10^{-10} m.

2(B)(b)(ii)**[2 marks]**

Calculate the speed of the electron in its orbit.

ANSWER

$$v \approx 2.2 \times 10^6 \text{ m s}^{-1}$$

RULES YOU NEED

- Speed = distance ÷ time = (distance per orbit) × (orbits per second)

WORKING — STEP BY STEP**1 Distance covered each second**It does 6.6×10^{15} orbits every second, each 3.33×10^{-10} m long:

$$v = 3.33 \times 10^{-10} \times 6.6 \times 10^{15}$$

2 Calculate $3.33 \times 6.6 = 21.98$, and $10^{-10} \times 10^{15} = 10^5$:

$$v = 21.98 \times 10^5 = 2.2 \times 10^6 \text{ m/s}$$

Marks	Awarded for
1	$v = 2\pi r \times f$.
1	$v \approx 2.2 \times 10^6 \text{ m s}^{-1}$.

2(B)(b)(iii)**[2 marks]**

Calculate the centripetal acceleration of the electron.

ANSWER

$$a_c \approx 9.1 \times 10^{22} \text{ m s}^{-2}$$

RULES YOU NEED

$$\bullet a_c = v^2 / r$$

WORKING — STEP BY STEP**1 Square the speed**

$$v^2 = (2.198 \times 10^6)^2 = 4.831 \times 10^{12} \text{ m}^2/\text{s}^2$$

2 Divide by r

$$a_c = 4.831 \times 10^{12} \div 5.3 \times 10^{-11} = 9.1 \times 10^{22} \text{ m s}^{-2}$$

Marks	Awarded for
1	$a_c = v^2/r$ with v from (ii).
1	$\approx 9.1 \times 10^{22} \text{ m s}^{-2}$ (own-figure accepted).

2(B)(b)(iv)**[2 marks]**

Calculate the centripetal force of interaction between the electron and proton.

ANSWER

$$F_c \approx 8.3 \times 10^{-8} \text{ N}$$

RULES YOU NEED

- $F = m a$

WORKING — STEP BY STEP**1 Multiply mass by acceleration**

$$F_c = 9.1 \times 10^{-31} \times 9.11 \times 10^{22}$$

2 Calculate

$$9.1 \times 9.11 = 82.9, \text{ and } 10^{-31} \times 10^{22} = 10^{-9};$$

$$F_c = 82.9 \times 10^{-9} = 8.3 \times 10^{-8} \text{ N}$$

Marks	Awarded for
1	$F = ma$ (or mv^2/r).
1	$\approx 8.3 \times 10^{-8} \text{ N}$.

Question 2(B) Total: 10 Marks

Question 3 · Option A

Temperature scales, thermal expansion, density of water — 10 marks

Given: The Spire: 120 m tall at 20 °C; stainless steel $\alpha = 1.5 \times 10^{-5} \text{ K}^{-1}$; winter day -10°C .

3(A)(a)

[3 marks]

Describe how the Celsius and Absolute (Kelvin) temperature scales are defined, in terms of the ice point and steam point of water.

ANSWER

Celsius: ice point 0 °C, steam point 100 °C. Kelvin: 273.15 K and 373.15 K, zero at absolute zero. $T_K = T_{^\circ\text{C}} + 273.15$.

RULES YOU NEED

- $T_K = T_{^\circ\text{C}} + 273.15$
- For temperature differences: $1 \text{ K} = 1^\circ\text{C}$

WORKING — STEP BY STEP

1 Celsius scale

The ice (freezing) point of water is set at **0 °C** and the steam (boiling) point at **100 °C**. The gap is split into 100 equal degrees.

2 Kelvin scale

The same size of degree, but shifted so zero is **absolute zero** (-273.15°C , where a gas's volume extrapolates to zero). So water freezes at **273.15 K** and boils at **373.15 K**.

3 Converting between them

$$T_K = T_{^\circ\text{C}} + 273.15$$

Only the zero point moves, so a change of 1°C is the same as a change of 1 K.

Marks	Awarded for
1	Celsius fixed points 0 °C and 100 °C.
1	Kelvin fixed points 273.15 K and 373.15 K, zero at absolute zero.
1	$T_K = T_{^\circ\text{C}} + 273.15$ / same size degree.

3(A)(b)**[1 mark]**

Write down an expression for the linear expansivity of a material, and give the SI unit for the coefficient of expansion.

ANSWER

$$\Delta L/L_0 = \alpha \Delta T, \text{ with } \alpha \text{ in } \text{K}^{-1} \text{ (equivalently } ^\circ\text{C}^{-1})$$

RULES YOU NEED

- Thermal expansivity (lectures, volume form): $\Delta V/V_0 = \beta \Delta T$

WORKING — STEP BY STEP**1 Apply the same idea to a length**

The fractional change in length is proportional to the temperature change:

$$\Delta L / L_0 = \alpha \Delta T$$

2 Work out the unit of α

$\Delta L/L_0$ is metres $\div$ metres, which has no unit. So $\alpha \times$ (kelvin) must have no unit, meaning α is in K^{-1} (per kelvin).

Note on the lecture material: The lecture notes give thermal expansivity in its volume form, $\Delta V/V_0 = \beta \Delta T$ (for a gas). The linear form is the same relation applied to length.

Marks	Awarded for
-------	-------------

1	Expression and unit K^{-1} .
---	---------------------------------------

3(A)(c)**[3 marks]**

What is the height of the Spire on a very cold winter day at $-10\text{ }^{\circ}\text{C}$?

ANSWER

Height $\approx 119.95\text{ m}$ (it shrinks by 5.4 cm)

RULES YOU NEED

- $\Delta L = \alpha L_0 \Delta T$
- $\Delta T = \text{final} - \text{initial}$

WORKING — STEP BY STEP**1 Temperature change**

$$\Delta T = -10 - 20 = -30\text{ }^{\circ}\text{C} = -30\text{ K}$$

Negative, because it gets colder.

2 Change in height

$$\Delta L = \alpha L_0 \Delta T = 1.5 \times 10^{-5} \times 120 \times (-30)$$

3 Calculate

$$1.5 \times 10^{-5} \times 120 = 1.8 \times 10^{-3}, \text{ then } \times (-30):$$

$$\Delta L = -0.054\text{ m (it shrinks by } 5.4\text{ cm)}$$

4 New height

$$L = 120 - 0.054 = 119.946\text{ m}$$

Marks	Awarded for
1	$\Delta T = -30\text{ K.}$
1	$\Delta L = \alpha L_0 \Delta T = -0.054\text{ m.}$
1	Final height $\approx 119.95\text{ m.}$

3(A)(d)**[2 marks]**

Draw a rough graph to show how density varies with temperature for (i) most liquids (ii) water, between 0 °C and 10 °C.

ANSWER

(i) Steadily decreasing line. (ii) Rises to a maximum at 4 °C, then falls.

RULES YOU NEED

- $\rho = m / V$: if a fixed mass expands, its density falls

WORKING — STEP BY STEP**1 Most liquids**

They expand steadily when heated. The mass stays the same while V grows, so $\rho = m/V$ **falls steadily** as temperature rises.

2 Water from 0 °C to 10 °C

Water is unusual: its density **rises** from 0 °C to a **maximum at 4 °C**, then falls as it warms further.

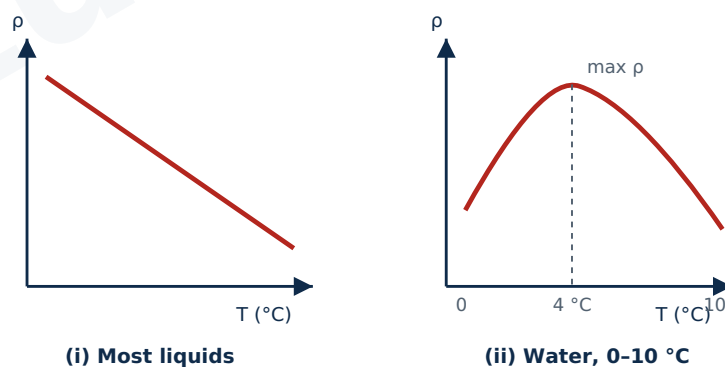
3 Sketch both

Fig. 1 — (i) Density falls steadily as temperature rises. (ii) Water's density peaks at 4 °C.

Note on the lecture material: Water's density maximum at 4 °C is **not covered in the PY1010 lecture notes**. This answer is included for completeness.

Marks	Awarded for
1	(i) Decreasing density with temperature.
1	(ii) Maximum at $\approx 4^{\circ}\text{C}$.

3(A)(e)**[1 mark]**

Using the second graph, explain why ponds freeze from the top down and not from the bottom up.

ANSWER

Water below 4 °C is less dense and rises, so the surface reaches 0 °C first and freezes; the densest (4 °C) water stays at the bottom

RULES YOU NEED

- Less dense fluid floats on denser fluid (buoyancy)

WORKING — STEP BY STEP**1 As the pond cools to 4 °C**

Water at 4 °C is the densest, so it sinks and collects at the bottom.

2 Below 4 °C

Colder water is **less dense**, so it floats to the top. The coldest water sits at the surface.

3 Freezing

The surface reaches 0 °C first and freezes. Ice is less dense than water, so it floats and forms a layer on top, while 4 °C water stays at the bottom.

Note on the lecture material: Relies on the same 4 °C behaviour of water, which is not covered in the lecture notes. The idea that less dense material floats does come from the buoyancy lectures.

Marks**Awarded for****1**

Links the 4 °C density maximum to colder, less dense water rising to the surface.

Question 3(A) Total: 10 Marks

Question 3 · Option B

Buoyancy and Archimedes' principle, a floating block of wood — 10 marks

Given: Wood density 700 kg m^{-3} ; volume 125 cm^3 ; floats in oil with 90% of its volume submerged.

3(B)(a)

[2 marks]

Briefly explain the concept of buoyancy and Archimedes' principle.

ANSWER

Buoyant force = weight of fluid displaced ($F_{\text{buoy}} = \rho_{\text{fl}} V_{\text{disp}} g$)

RULES YOU NEED

- $F_{\text{buoy}} = \text{weight of fluid displaced} = \rho_{\text{fluid}} V_{\text{displaced}} g$

WORKING — STEP BY STEP

1 Buoyancy

A fluid pushes up on anything placed in it, because the pressure in a fluid is bigger at greater depth: the bottom of the object is pushed up harder than the top is pushed down.

2 Archimedes' principle

The upward (buoyant) force equals the **weight of the fluid displaced** by the object.

$$F_{\text{buoy}} = \rho_{\text{fluid}} V_{\text{displaced}} g$$

3 Floating or sinking

If the buoyant force can balance the object's weight, it floats; if not, it sinks.

Marks	Awarded for
1	Buoyancy as an upward force from the fluid.
1	Archimedes' principle: equals the weight of fluid displaced.

3(B)(b)(i)**[2 marks]**

Calculate the mass of the wood.

ANSWER

$m = 0.0875 \text{ kg (87.5 g)}$

RULES YOU NEED

- $m = \rho V$
- $1 \text{ cm}^3 = 10^{-6} \text{ m}^3$

WORKING — STEP BY STEP**1 Convert the volume**

$$V = 125 \text{ cm}^3 = 125 \times 10^{-6} \text{ m}^3 = 1.25 \times 10^{-4} \text{ m}^3$$

2 Multiply by the density

$$m = 700 \times 1.25 \times 10^{-4} = 0.0875 \text{ kg}$$

Marks	Awarded for
1	Volume in m^3 , $m = \rho V$.
1	0.0875 kg.

3(B)(b)(ii)**[4 marks]**

Calculate the density of the oil.

ANSWER

$$\rho_{\text{oil}} \approx 7.8 \times 10^2 \text{ kg m}^{-3} \text{ (778 kg m}^{-3}\text{)}$$

RULES YOU NEED

- Floating in equilibrium: buoyant force = weight
- $F_{\text{buoy}} = \rho_{\text{oil}} V_{\text{submerged}} g$; weight = $\rho_{\text{wood}} V g$

WORKING — STEP BY STEP**1 The block floats, so the forces balance**

$$F_{\text{buoy}} = \text{weight}$$

2 Write each side outOnly 90% of the block is under the oil, so $V_{\text{submerged}} = 0.9V$:

$$\rho_{\text{oil}} \times 0.9 V \times g = \rho_{\text{wood}} \times V \times g$$

3 Cancel V and g

$$0.9 \rho_{\text{oil}} = \rho_{\text{wood}} \Rightarrow \rho_{\text{oil}} = \rho_{\text{wood}} / 0.9$$

4 Calculate

$$\rho_{\text{oil}} = 700 \div 0.9 = 778 \text{ kg m}^{-3}$$

Sense check: the wood floats with most of it under, so the oil is only a bit denser than the wood ✓

Marks	Awarded for
1	Floating: buoyant force = weight.
1	Displaced volume = 0.9V.
1	Rearranged for ρ_{oil} .
1	$\approx 778 \text{ kg m}^{-3}$.

3(B)(b)(iii)**[2 marks]**

When you pour the oil into water the two do not mix. Would you expect the oil to be above or below the water? Briefly justify.

ANSWER

Above the water, because oil is less dense ($778 < 1000 \text{ kg m}^{-3}$)

RULES YOU NEED

- A less dense substance floats on a denser one

WORKING — STEP BY STEP**1 Compare the densities**

Oil $\approx 778 \text{ kg m}^{-3}$; water = 1000 kg m^{-3} . The oil is less dense.

2 Apply buoyancy

Any bit of oil in the water displaces water that weighs more than the oil itself, so the buoyant force pushes it up. The oil ends up **on top**.

Marks	Awarded for
1	Above.
1	Justified by comparing densities / buoyancy.

Question 3(B) Total: 10 Marks

Question 4 · Option A

Frequency, types of wave, $v = f\lambda$, seismic wave — 10 marks

Given: Seismic wave: frequency 10 Hz, wavelength 500 m, travel time 100 s.

4(A)(a)

[2 marks]

What is meant by the frequency of a wave? Give its SI unit.

ANSWER

f = number of oscillations per second ($f = 1/P$); unit hertz, Hz = s^{-1}

RULES YOU NEED

- $f = 1 / P$ (P = period, time for one oscillation)
- $1 \text{ Hz} = 1 \text{ s}^{-1}$

WORKING — STEP BY STEP

1 Definition

The frequency f is the number of complete oscillations (e.g. crests passing a point) every second.

2 Link to the period

The period P is the time for one oscillation, so $f = 1/P$.

3 Unit

Oscillations per second: s^{-1} , called the **hertz (Hz)**.

Marks	Awarded for
1	Definition (oscillations per second / 1/period).
1	Hz (s^{-1}).

4(A)(b)**[4 marks]**

Describe the two main types of wave. Give four examples of wave motion.

ANSWER

Transverse ($\perp$ to propagation) and longitudinal ($\parallel$ to propagation); examples: water, seismic, sound, light

RULES YOU NEED

- A wave transfers energy without net motion of the medium.

WORKING — STEP BY STEP**1 Transverse waves**

The medium moves **perpendicular** to the direction the wave travels. E.g. a wave on a rope, water waves, light.

2 Longitudinal waves

The medium vibrates **parallel** to the direction the wave travels, forming compressions and rarefactions. E.g. sound in air.

3 Four examples of wave motion

Water waves, seismic waves, sound, light.

Marks	Awarded for
1	Transverse described correctly.
1	Longitudinal described correctly.
2	Four valid examples (½ mark each).

4(A)(c)**[1 mark]**

Write down the expression relating the velocity, frequency and wavelength of a wave.

ANSWER

$$v = f\lambda$$

RULES YOU NEED

- In one second, f crests pass, each λ apart, so the wave moves $f\lambda$ metres.

WORKING — STEP BY STEP**1 The formula**

$$v = f\lambda$$

v in m s^{-1} , f in Hz, λ in m. It holds for all waves.

Marks**Awarded for****1** $v = f\lambda$.

4(A)(d)**[3 marks]**

A seismic wave has frequency 10 Hz and wavelength 500 m. If it took 100 s to reach the geologist, how far away was the earthquake?

ANSWER

$$d = 5 \times 10^5 \text{ m (500 km)}$$

RULES YOU NEED

- $v = f \lambda$
- Distance = speed $\times$ time

WORKING — STEP BY STEP**1 Find the wave speed**

$$v = f \lambda = 10 \text{ Hz} \times 500 \text{ m} = 5000 \text{ m/s}$$

2 It travels in a straight line at constant speed

$$d = v t = 5000 \text{ m/s} \times 100 \text{ s} = 500\,000 \text{ m}$$

3 Convert to km

$$d = 5 \times 10^5 \text{ m} = 500 \text{ km}$$

Marks	Awarded for
1	$v = f\lambda = 5000 \text{ m s}^{-1}$.
1	$d = vt$.
1	500 km.

Question 4(A) Total: 10 Marks

Question 4 · Option B

Resistance, current and power; charge flowing through a resistor — 10 marks

Given: A charge of 2 C flows through a 10 Ω resistance in 5 s.

4(B)(a)

[3 marks]

Define (i) electric resistance, (ii) electric current, and (iii) electric power.

ANSWER

$$R = V/I \text{ (}\Omega\text{); } I = Q/t \text{ (A); } P = E/t = IV \text{ (W)}$$

RULES YOU NEED

- $V = IR$
- $I = Q/t$
- $P = E/t = IV$

WORKING — STEP BY STEP

1 Resistance

The voltage across a conductor divided by the current through it, $R = V/I$. It measures how strongly the conductor opposes the flow of charge. Unit: ohm (Ω).

2 Current

The rate of flow of charge, $I = Q/t$. Unit: ampere, $A = C s^{-1}$.

3 Power

The rate at which electrical energy is used, $P = E/t = IV$. Unit: watt (W) = $J s^{-1}$.

Marks	Awarded for
1	Resistance $R = V/I$.
1	Current $I = Q/t$.
1	Power = energy per time, $P = IV$.

4(B)(b)(i)**[1 mark]**

What is the current through the resistance?

ANSWER

$I = 0.4 \text{ A}$

RULES YOU NEED

• $I = Q / t$

WORKING — STEP BY STEP**1 Charge per second**

$$I = 2 \text{ C} \div 5 \text{ s} = 0.4 \text{ A}$$

Marks**Awarded for****1**

0.4 A.

4(B)(b)(ii)**[2 marks]**

How much work is done on the charge?

ANSWER

$W = 8 \text{ J}$

RULES YOU NEED

- $V = IR$
- Work done moving charge q through voltage V : $W = Vq$

WORKING — STEP BY STEP**1 Voltage across the resistor**

$$V = IR = 0.4 \text{ A} \times 10 \Omega = 4 \text{ V}$$

2 Work done on the 2 C

Each coulomb gets 4 J, and there are 2 C:

$$W = Vq = 4 \text{ V} \times 2 \text{ C} = 8 \text{ J}$$

Marks	Awarded for
1	$V = IR = 4 \text{ V}.$
1	$W = Vq = 8 \text{ J}.$

4(B)(b)(iii)**[2 marks]**

At what rate is power dissipated by the resistor?

ANSWER

$P = 1.6 \text{ W}$

RULES YOU NEED

- $P = W / t$
- $P = I^2 R$

WORKING — STEP BY STEP**1 Energy per second**

$$P = W / t = 8 \text{ J} \div 5 \text{ s} = 1.6 \text{ W}$$

2 Check a second way

$$P = I^2 R = 0.4^2 \times 10 = 0.16 \times 10 = 1.6 \text{ W} \checkmark$$

Marks	Awarded for
1	$P = W/t$ (or I^2R , IV).
1	1.6 W.

4(B)(b)(iv)**[2 marks]**

A second, identical resistor is connected in series. What is the total voltage drop across both resistors if the current stays the same?

ANSWER

$$V_{\text{total}} = 8 \text{ V}$$

RULES YOU NEED

- $V = I R$ for each resistor
- $V =$ work per unit charge, so the voltages along one path add up

WORKING — STEP BY STEP**1 Same current through both**

In series there is only one path, so 0.4 A flows through each 10 Ω resistor.

2 Voltage across each

$$V = I R = 0.4 \times 10 = 4 \text{ V each}$$

3 Add them

Each coulomb loses 4 J in the first resistor and 4 J in the second:

$$V_{\text{total}} = 4 + 4 = 8 \text{ V}$$

4 Check

Total resistance 20 Ω : $V = 0.4 \times 20 = 8 \text{ V} \checkmark$

Marks	Awarded for
1	Same current through both; 4 V across each (or $R_{\text{total}} = 20 \Omega$).
1	8 V.

Question 4(B) Total: 10 Marks

Question 5 · Option A

Lorentz force on a wire, the electric motor loop, field of a current loop — 10 marks

Given: Square loop in the field between N and S poles (B from N to S, left $\rightarrow$ right); loop can rotate about the axis shown.

5(A)(a)

[4 marks]

Write down the formula for the magnitude of the Lorentz force on a current-carrying wire in an external field B , with the current perpendicular to B . Draw a diagram relating the directions of I , B and the force.

ANSWER

$F = BIl$, with F perpendicular to both I and B (Left Hand Rule)

RULES YOU NEED

- $F = BIl$ ($I \perp B$)
- Force $\perp$ to both I and B
- Left Hand Rule: First finger = Field, seCond finger = Current, thuMb = Motion (force)

WORKING — STEP BY STEP

1 The formula

$$F = BIl$$

F = force (N), B = magnetic field (T), I = current (A), ℓ = length of wire inside the field (m).

2 Direction: use the Left Hand Rule

Hold your first three fingers at right angles. First finger along B , second finger along I ; your thumb gives F .

3 Draw the diagram

Example: current to the right, B into the page, force up the page. All three are at right angles to each other.

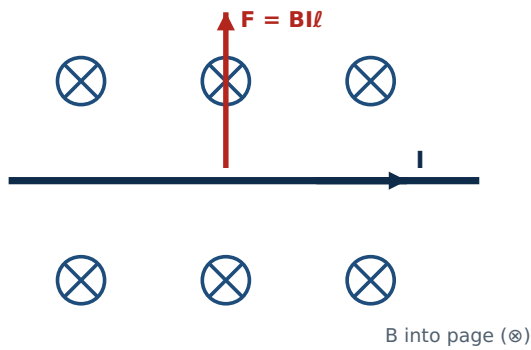


Fig. 2 — Current to the right in a field into the page: the force points up the page.

Marks	Awarded for
1	$F = BI\ell$.
1	Terms defined (B , I , ℓ).
2	Diagram with I , B and F mutually perpendicular in a consistent (left-hand-rule) arrangement.

5(A)(b)(i)**[2 marks]**

The loop's plane is parallel to B and I flows as shown. Does the loop rotate (A) clockwise, (B) anticlockwise, or (C) remain fixed?

ANSWER**(B) Anticlockwise****RULES YOU NEED**

- $F = BIl$ only for the parts of the wire perpendicular to B; parts parallel to B feel no force
- Left Hand Rule for each side

WORKING — STEP BY STEP**1 Which sides feel a force?**

B runs from N to S (left to right). The two sides of the loop that lie along B feel **no** force. Only the two sides running along the axis direction (perpendicular to B) feel a force.

2 Current direction in those two sides

Follow the current in from the leads: on the **left** side it flows **into** the page (away from you); on the **right** side it flows **out of** the page (towards you).

3 Left Hand Rule on each side

Left side: B right, I into page → force **down**. Right side: B right, I out of page → force **up**.

4 What do the two forces do?

Equal and opposite forces on opposite sides of the axis twist the loop: the right side goes up and the left side goes down.

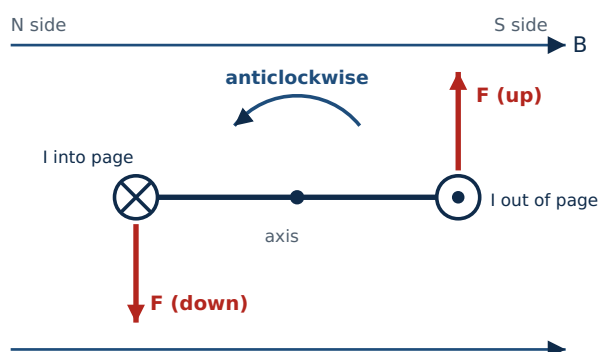


Fig. 3 — View along the axis: left side pushed down, right side up, so the loop turns anticlockwise.

5 Conclusion

Right up, left down, seen looking along the axis, is **anticlockwise**. This is how an electric motor works.

Marks	Awarded for
1	Forces only on the sides perpendicular to B, in opposite directions (up/down).
1	(B) anticlockwise.

5(A)(b)(ii)**[2 marks]**

If I flows in the opposite direction around the loop and the direction of B is reversed, which of (A), (B), (C) now holds?

ANSWER

(B) Anticlockwise, the same as before

RULES YOU NEED

- Reversing I reverses the force; reversing B reverses the force.

WORKING — STEP BY STEP**1 Reverse the current only**

Every force would flip: the loop would turn clockwise.

2 Now also reverse B

Every force flips back again.

3 Conclusion

Two flips cancel, so the forces are the same as in (i): **anticlockwise**.

Marks	Awarded for
1	Recognises that the two reversals cancel.
1	(B) anticlockwise.

5(A)(c)**[2 marks]**

Sketch the magnetic field generated by the current in a small circular loop of wire, and relate it to the field of one other type of magnetic object.

ANSWER

A dipole field, like that of a bar magnet (N on one face of the loop, S on the other)

RULES YOU NEED

- Right-hand rule: thumb along I , fingers curl in the direction of B

WORKING — STEP BY STEP**1 Look at the loop edge-on**

You see two cross-sections: current coming out of the page on one side ($\odot$) and going into it on the other ($\otimes$).

2 Field around each side

By the right-hand rule, B circles each side, in opposite senses. Between the two sides, both fields point the **same way** through the middle of the loop.

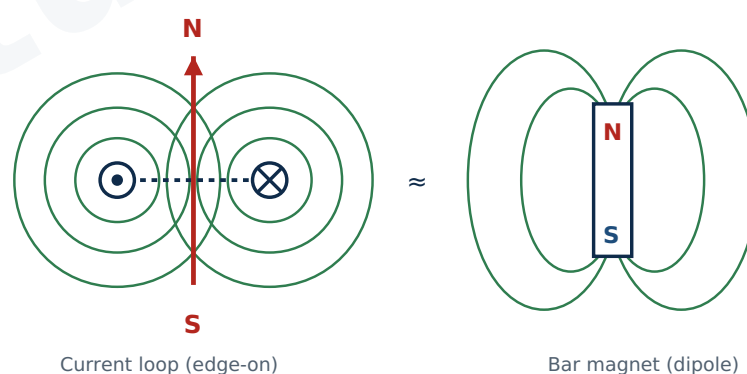
3 Sketch the pattern

Fig. 4 — Field lines pass the same way through the loop and loop back outside: a dipole, like a bar magnet.

4 Relate it to another magnet

This is a **dipole** field, the same shape as a **bar magnet**: one face of the loop acts as N, the other as S. (The same idea explains the Earth's field.)

Marks	Awarded for
1	Sketch: field lines through the centre of the loop, returning outside.
1	Related to a bar magnet (or Earth / dipole).

Question 5(A) Total: 10 Marks

Study-Uni

Question 5 · Option B

Faraday's law of induction, magnetic flux, induced voltage and current — 10 marks

Given: Circular loop of area 0.3 m^2 , held horizontal; a vertical B increases from 0 to 2 T in 3 s; wire resistance 2Ω .

5(B)(a)

[1 mark]

Write down Faraday's Law of induction, explaining each of the terms.

ANSWER

$V = \Delta\Phi/\Delta t$: induced voltage = rate of change of magnetic flux

RULES YOU NEED

- Magnetic flux $\Phi = B A$

WORKING — STEP BY STEP

1 The law

$$V = \Delta\Phi / \Delta t$$

2 The terms

V = induced voltage (V). $\Delta\Phi$ = change in magnetic flux through the loop (Wb). Δt = time taken for the change (s). A voltage is induced only **while** the flux is changing.

Marks	Awarded for
1	Law with terms explained.

5(B)(b)**[1 mark]**

Give the SI unit for magnetic flux.

ANSWER

The weber, Wb (= T m²)

RULES YOU NEED

- $\Phi = B A$

WORKING — STEP BY STEP**1 From the formula**

$\Phi = BA$, so the unit is tesla $\times$ m² = T m², called the **weber (Wb)**.

Marks**Awarded for****1**Wb (or T m²).

5(B)(c)(i)**[2 marks]**

Between $t = 0$ and $t = 3$ s, what is the increase in the magnetic flux through the loop?

ANSWER

$$\Delta\Phi = 0.6 \text{ Wb}$$

RULES YOU NEED

- $\Phi = B A$ (B perpendicular to the loop)

WORKING — STEP BY STEP**1 Is B perpendicular to the loop?**

The loop is horizontal and B is vertical, so yes: all the field lines pass straight through, and $\Phi = BA$.

2 Flux at the start and end

$$\Phi_{\text{start}} = 0 \times 0.3 = 0 \text{ Wb}$$

$$\Phi_{\text{end}} = 2 \text{ T} \times 0.3 \text{ m}^2 = 0.6 \text{ Wb}$$

3 Increase

$$\Delta\Phi = 0.6 - 0 = 0.6 \text{ Wb}$$

Marks	Awarded for
1	$\Phi = BA$ with $B \perp$ loop.
1	0.6 Wb.

5(B)(c)(ii)**[2 marks]**

What is the induced voltage?

ANSWER

$V = 0.2 \text{ V}$

RULES YOU NEED

- Faraday: $V = \Delta\Phi / \Delta t$

WORKING — STEP BY STEP**1 Apply Faraday's law**

$$V = 0.6 \text{ Wb} \div 3 \text{ s} = 0.2 \text{ V}$$

Marks	Awarded for
1	Faraday's law applied.
1	0.2 V.

5(B)(c)(iii)**[2 marks]**

If the resistance of the wire is $2\ \Omega$, what current flows?

ANSWER

$I = 0.1\ \text{A}$

RULES YOU NEED

- $V = I R$, so $I = V / R$

WORKING — STEP BY STEP**1 Apply $I = V/R$**

$$I = 0.2\ \text{V} \div 2\ \Omega = 0.1\ \text{A}$$

Marks	Awarded for
1	$I = V/R$.
1	0.1 A.

5(B)(c)(iv)**[2 marks]**

For $t > 3$ s, B stays at 2 T. Does the induced current increase, stay the same, or drop to zero?

ANSWER**It drops to zero: the flux is no longer changing****RULES YOU NEED**

- A voltage is induced only while the flux is changing

WORKING — STEP BY STEP**1 Is the flux still changing?**

B is fixed at 2 T and the loop doesn't move, so Φ stays at 0.6 Wb: $\Delta\Phi = 0$.

2 Apply Faraday's law

$$V = 0 / \Delta t = 0, \text{ so } I = V/R = 0$$

3 Conclusion

The induced current **drops to zero**.

Marks	Awarded for
1	Drops to zero.
1	Reason: no change in flux, so no induced V .

Question 5(B) Total: 10 Marks

Question 6 · Option A

Lens sign conventions, magnification, the thin lens equation — 10 marks

Given: Converging lens, $f = 30\text{ cm}$; real image three times further from the lens than the object.

6(A)(a)(i)

[1 mark]

The focal length f of a concave lens is (A) always positive, (B) always negative, or (C) can be either.

ANSWER

(B) Always negative

RULES YOU NEED

- Thin lens sign rule: $f < 0$ for a concave lens, $f > 0$ for a convex lens — always

WORKING — STEP BY STEP

1 Apply the rule

A concave (diverging) lens always has $f < 0$.

Marks	Awarded for
-------	-------------

1	(B).
---	------

6(A)(a)(ii)**[1 mark]**

The focal length f of a convex lens is (A) always positive, (B) always negative, or (C) can be either.

ANSWER**(A) Always positive****RULES YOU NEED**

- $f > 0$ for a convex lens — always

WORKING — STEP BY STEP**1 Apply the rule**

A convex (converging) lens always has $f > 0$.

Marks	Awarded for
-------	-------------

1	(A).
---	------

6(A)(a)(iii)**[1 mark]**

The image distance v of a convex lens is (A) always positive, (B) always negative, or (C) can be either.

ANSWER**(C) Can be either****RULES YOU NEED**

- $v > 0$ for a real image, $v < 0$ for a virtual image

WORKING — STEP BY STEP**1 Object beyond f**

The rays really cross on the far side: a real image, $v > 0$.

2 Object closer than f (magnifying glass)

A virtual image forms, $v < 0$. Lecture example: $u = 10$ cm, $f = 20$ cm gives $v = -20$ cm.

3 Conclusion

Both are possible: v can be either sign.

Marks	Awarded for
-------	-------------

1	(C).
---	------

6(A)(a)(iv)**[1 mark]**

The magnification of a convex lens is (A) always greater than one, (B) equals v/u , or (C) both.

ANSWER**(B) Equals v/u** **RULES YOU NEED**

- $M = h_i/h_o = v/u$ (similar triangles)

WORKING — STEP BY STEP**1 Is $M = v/u$?**

Yes, from similar triangles through the centre of the lens.

2 Is M always greater than 1?

No. Lecture example: object at $u = 10$ cm, $f = 4$ cm gives $v = 6.67$ cm and $M = 0.66$, a smaller image.

3 Conclusion

Only (B) is always true.

Marks	Awarded for
1	(B).

6(A)(b)(i)**[1 mark]**

Is the image erect or inverted?

ANSWER**Inverted****RULES YOU NEED**

- A real image from a converging lens is inverted

WORKING — STEP BY STEP**1 Real image**

The image is real: the rays actually cross on the far side of the lens.

2 Ray tracing

A ray from the top of the object through the centre of the lens is undeviated, so it crosses the axis and lands below it. The image is **inverted**.

Marks Awarded for**1** Inverted.

6(A)(b)(ii)**[2 marks]**

What is the magnification of the object?

ANSWER**M = 3 (image three times the size of the object, inverted)****RULES YOU NEED**

- $M = v / u$

WORKING — STEP BY STEP**1 Use the given relation**The image is three times as far from the lens as the object: $v = 3u$.**2 Substitute**

$$M = v / u = 3u / u = 3$$

Marks	Awarded for
1	$M = v/u$.
1	$M = 3$.

6(A)(b)(iii)**[3 marks]**

What are the distances of the object and the image from the lens?

ANSWER**Object at $u = 40$ cm; image at $v = 120$ cm****RULES YOU NEED**

- Thin lens equation: $1/u + 1/v = 1/f$
- Convex lens: $f = +30$ cm; real image: $v > 0$

WORKING — STEP BY STEP**1 Substitute $v = 3u$**

$$1/u + 1/(3u) = 1/30$$

2 Combine the fractions

$$1/u = 3/(3u), \text{ so } 3/(3u) + 1/(3u) = 4/(3u):$$

$$4 / (3u) = 1/30$$

3 Solve for u

$$3u = 4 \times 30 = 120 \Rightarrow u = 40 \text{ cm}$$

4 Find v

$$v = 3u = 120 \text{ cm}$$

5 Check in the equation

$$1/40 + 1/120 = 3/120 + 1/120 = 4/120 = 1/30 \checkmark$$

Marks	Awarded for
1	Thin lens equation with $v = 3u$.
1	$u = 40$ cm.
1	$v = 120$ cm.

Question 6(A) Total: 10 Marks

Question 6 · Option B

Snell's law, refraction at the water surface seen by a diver — 10 marks

Given: A submerged diver sees the Sun 35° from the vertical; $n_{\text{water}} = 1.33$; $n_{\text{air}} = 1$.

6(B)(a)

[3 marks]

Write down Snell's Law of Refraction and, with a diagram, briefly explain all the terms.

ANSWER

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \text{ (angles measured from the normal)}$$

RULES YOU NEED

- $n_1 \sin \theta_1 = n_2 \sin \theta_2$
- $n = c / (\text{speed of light in the medium})$

WORKING — STEP BY STEP

1 The law

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

2 Draw the set-up

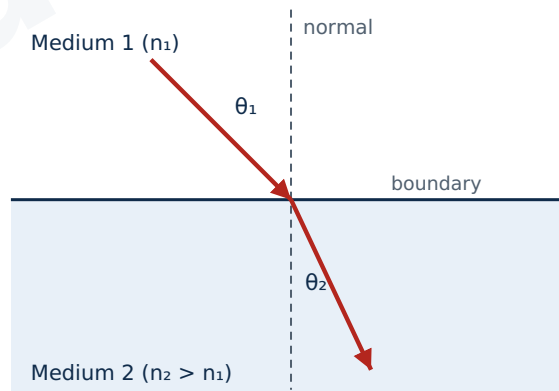


Fig. 5 — A ray passing into an optically denser medium bends towards the normal.

3 Explain the terms

n_1 , n_2 : refractive indices of the two media, $n = c / (\text{speed of light in the medium})$: air ≈ 1 , water ≈ 1.33 .

θ_1 : angle of incidence; θ_2 : angle of refraction. Both are measured from the **normal** (the line at 90° to the boundary). The incident ray, refracted ray and normal all lie in one plane.

Marks	Awarded for
1	Snell's law stated.
1	n and θ terms explained (angles from the normal).
1	Labelled diagram.

Study-Uni

6(B)(b)(i)**[3 marks]**

Draw a diagram of the path of a ray from the Sun, noting the angles of incidence as observed by the diver and her friend on the shore.

ANSWER

Ray bends towards the vertical on entering the water; $\theta_{\text{air}} > \theta_{\text{water}} = 35^\circ$

RULES YOU NEED

- Going into a denser medium, the ray bends towards the normal

WORKING — STEP BY STEP**1 Where is the normal?**

The water surface is flat and horizontal, so the normal is **vertical**. “From the vertical” means “from the normal”.

2 Which angle is bigger?

Light goes from air ($n = 1$) into water ($n = 1.33$), a denser medium, so it bends towards the normal. The angle in air (seen by the friend) is **bigger** than the 35° in water (seen by the diver).

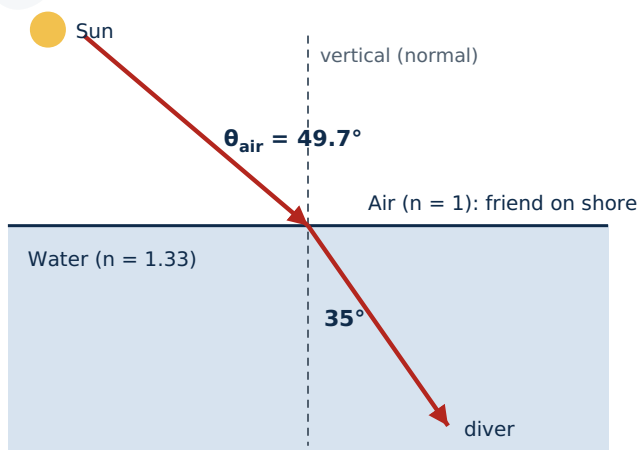
3 Draw it

Fig. 6 — The Sun's ray bends towards the normal entering the water: 49.7° in air becomes 35° in water.

Marks	Awarded for
1	Horizontal surface with vertical normal drawn.
1	Ray bending towards the normal in the water.
1	Angles labelled: θ_{air} (friend) and 35° (diver).

6(B)(b)(ii)**[4 marks]**

At what angle from the vertical does the diver's friend see the Sun? ($n_{\text{water}} = 1.33$)

ANSWER

The friend sees the Sun $\approx 50^\circ$ (49.7°) from the vertical

RULES YOU NEED

$$n_{\text{air}} \sin \theta_{\text{air}} = n_{\text{water}} \sin \theta_{\text{water}}$$

WORKING — STEP BY STEP**1 Set up Snell's law at the surface**

$$1 \times \sin \theta_{\text{air}} = 1.33 \times \sin 35^\circ$$

2 Find $\sin 35^\circ$

$$\sin 35^\circ = 0.5736$$

3 Multiply

$$\sin \theta_{\text{air}} = 1.33 \times 0.5736 = 0.7629$$

4 Take the inverse sine

$$\theta_{\text{air}} = \sin^{-1}(0.7629) = 49.7^\circ$$

5 Sense check

$49.7^\circ > 35^\circ$, which matches the ray bending towards the normal as it enters the water ✓

Marks	Awarded for
1	Snell's law set up with the correct media.
1	35° assigned to the water side.
1	$\sin \theta_{\text{air}} = 0.763$.
1	$\theta_{\text{air}} \approx 49.7^\circ$.

Question 6(B) Total: 10 Marks

Study-Uni